BINARY SEARCH - C PROGRAM

1. PROGRAM CODE

```c
#include <stdio.h>

int main()
{
    int arr[100], n, key;
    int low, high, mid, found = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d sorted elements:\n", n);
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter the item to search: ");
    scanf("%d", &key);

    low = 0;
    high = n - 1;

    while (low <= high)
    {
        mid = (low + high) / 2;

        if (arr[mid] == key)
        {
            found = 1;
            printf("Item found at position: %d\n", mid + 1);
            printf("Index: %d\n", mid);
            break;
        }
        else if (arr[mid] < key)
        {
            low = mid + 1;
        }
        else
        {
            high = mid - 1;
        }
    }

    if (found == 0)
    {
        printf("Item not found.\n");
    }

    return 0;
}
```

2. ACTUAL PROGRAM OUTPUT

Representative full run (captured stdout)

```
Enter the number of elements: Enter 7 sorted elements:
Enter the item to search: Item found at position: 5
Index: 4
```